Pre-selection: How the Tool Chooses the Indicators

Six screens, one race — with an application to Trinidad and Tobago's Real GDP Growth

Jose Luis Saboin, PhD
Inter-American Development Bank

GDP Growth Nowcasting Workshop: Trinidad and Tobago

October 27, 2026

"Four hundred candidates, forty-one quarters: somebody has to make a short list."

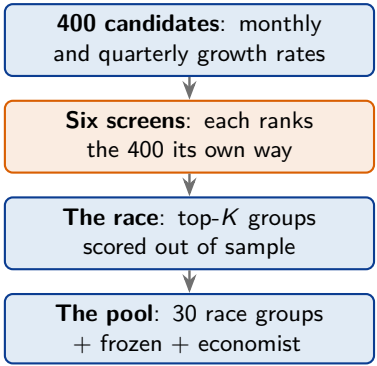
Part I — The problem

Four hundred candidates, forty-one quarters

The Trinidad and Tobago panel of October 2026: **638 series**, of which **400 pass the data rules** (335 monthly). The production training window runs from 2013-Q1 to 2023-Q1: **41 quarters**.

More candidates than observations, ten to one. Three things go wrong if every candidate is used at once:

- **A regression cannot be estimated.** With 400 regressors and 41 observations there are infinitely many exact fits and no way to choose.
- **One factor from everything is mostly noise.** The first principal component of 400 growth rates follows the most volatile series, not the economy.
- **Trees and networks memorise.** Give a boosted tree or an LSTM 400 inputs and 41 examples and it reproduces the sample, then fails outside it.



The screens are cheap (seconds). The estimators that follow are expensive (hours). Pre-selection is where the budget is decided.

The committee analogy, used throughout

Think of a **hiring committee with six interviewers**. Each interviewer meets all 400 applicants and hands in a ranked list. The interviewers disagree, because each asks a different question:

Screen	The question it asks	Style
SIS	Who moves with GDP?	one applicant at a time
t-stat	Who adds to what GDP's own past already says?	one at a time, with controls
OCMT	Same, but with a bar that rises with the number of applicants	one at a time, two rounds
LARS	Whom would a regression hire first, if they compete?	competitive
SHAP	Whom does a boosted-tree model actually use?	competitive, non-linear
iBMA	In how many good teams does the applicant appear?	competitive, model averaging

No interviewer hires. Each list goes to a **tryout** (the race, Part III): short lists of the top 5, top 6, . . . , top 99 of each ranking are tested on quarters the ranking never saw. The lists that survive become the groups the estimators are trained on.

The example used in every screen: five indicators, twelve quarters

Five indicators from the tool's frozen groups (the lists behind the watchlist), twelve quarters, growth rates as the tool uses them: quarterly series in % q/q, monthly series as the quarter mean of the % m/m changes, all X-13 adjusted. y = SA RGDP growth (% q/q).

Quarter	y: RGDP	Jobs searches UGTR1001	Fuel imports XBOP1115	Real estate credit MCRD7000	Private credit MCRD5000	ATM withdrawals RPAY5006
2016-Q1	−1.33	−3.56	13.17	0.24	0.08	0.25
2016-Q2	−4.50	1.75	7.80	−0.04	−0.05	−0.49
2016-Q3	−3.67	5.95	13.96	0.14	0.16	2.07
2016-Q4	3.72	−6.55	5.89	0.17	0.12	−2.63
2017-Q1	−4.16	3.20	−8.68	0.34	0.00	3.65
2017-Q2	0.24	−4.08	−15.25	0.51	0.36	−2.66
2017-Q3	−0.34	3.20	13.29	0.86	0.45	−0.53
2017-Q4	0.18	−0.91	31.32	0.47	0.43	0.39
2018-Q1	−0.19	0.26	−2.96	0.39	0.52	−1.60
2018-Q2	0.79	0.64	12.19	0.86	0.20	0.96
2018-Q3	−2.83	0.05	−1.91	0.36	−0.04	1.28
2018-Q4	−1.98	−0.81	−64.76	0.21	−0.08	1.27

Jobs searches: Google Trends index for "jobs" (TT); fuel imports in US$; credit and ATM series in real TT$. Twelve quarters is far too few for a real ranking, and enough to do every screen by hand.

Part II — The six screens

Screen 1 — SIS: who moves with GDP?

The question it asks: *Taken alone, how strongly does the candidate co-move with GDP growth?*

The rule. Compute the correlation of each candidate with y over the training window, take its absolute value, sort. That is the whole screen (Fan and Lv, 2008: *sure independence screening*). Rankings are nested: the top 20 are the first 20 of the top 40.

Candidate	corr with y	$\lvert\text{corr}\rvert$
Jobs searches	−0.70	**0.70**
ATM withdrawals	−0.69	**0.69**
Private credit	0.51	0.51
Real estate credit	0.41	0.41
Fuel imports	0.16	0.16

- The sign is irrelevant for the ranking: more searches for jobs go with weaker growth, and SIS keeps the series because the link is strong, not because it is positive.
- Jobs searches and ATM withdrawals are nearly tied at the top. Fuel imports, the only series whose swings are tens of percent, comes last: size of movement is not co-movement.
- Cost: one correlation per candidate, 400 numbers.

$$\text{SIS}(x_j) = \left| \frac{\sum_t (x_{jt} - \bar{x}_j)(y_t - \bar{y})}{\sqrt{\sum_t (x_{jt} - \bar{x}_j)^2 \sum_t (y_t - \bar{y})^2}} \right|$$

What SIS cannot see: the applicants are not independent

Correlation between the five candidates, same twelve quarters:

	Jobs	Fuel	RE cr.	Pr. cr.	ATM
Jobs	1	.13	.09	−.02	**.66**
Fuel	.13	1	.21	.40	−.09
RE cr.	.09	.21	1	**.58**	−.05
Pr. cr.	−.02	.40	.58	1	−.48
ATM	.66	−.09	−.05	−.48	1

- SIS interviews each applicant alone. Its two favourites, jobs searches and ATM withdrawals, are correlated at 0.66: they carry much the same news. A top-2 list of SIS is partly one piece of information counted twice.
- The two credit series overlap too (0.58).
- This is the weakness of every *one-at-a-time* screen, and the reason the committee has competitive interviewers as well (LARS, SHAP, iBMA): they let the applicants compete and keep the one that adds something.
- The tool also attacks redundancy before any ranking (Part III): identities are dropped and a candidate with correlation above 0.8 with one already kept is removed from the ranking.

Screen 2 — t-stat: who adds to what GDP's past already says?

The question it asks: *Once GDP's own two lags are in the regression, does the candidate still explain growth?*

The rule. One small regression per candidate: GDP growth on a constant, its own two lags and the candidate. Rank by the absolute t-statistic of the candidate. A series that only repeats what last quarter's GDP says earns a small t (Bai and Ng, 2008; in macro, Jurado, Ludvigson and Ng, 2015).

- **Why the lags.** Growth is persistent; many series co-move with GDP simply because both carry last quarter's momentum. The lags remove that shared part first.
- **Why HAC standard errors.** Quarterly residuals are serially correlated; naive standard errors would inflate every t. The tool uses Newey-West with four lags.
- **Still one at a time.** The regressions are separate, so two correlated candidates can both score high. Competition comes later.

$$y_t = c + b_1\, y_{t-1} + b_2\, y_{t-2} + g_j\, x_{jt} + e_t, \qquad \text{score}(x_j) = \left| \frac{\hat{g}_j}{\text{se}_{\text{HAC}}(\hat{g}_j)} \right|$$

Screen 2 by hand: five regressions

Twelve quarters, two GDP lags (the lags of 2016-Q1 reach back to 2015-Q3). The lags alone explain little here: $R^2 = 0.11$, coefficients -0.35 and -0.15.

Candidate x_j	$\hat{g}_j$	se (HAC)	$\lvert t \rvert$
Jobs searches	-0.489	0.054	**9.1**
ATM withdrawals	-0.910	0.171	5.3
Private credit	5.876	1.238	4.8
Real estate credit	5.384	1.479	3.6
Fuel imports	0.019	0.018	1.1

- **Same top, different order below it.** Jobs searches leads as in SIS, but ATM withdrawals falls behind once the lags are in: part of its co-movement with GDP was momentum.
- The credit series rise. Their coefficients look large only because credit moves by fractions of a percent while GDP moves by percents; the t-statistic is scale-free, which is why the screen ranks on t and not on $\hat{g}$.
- Fuel imports stays last. Its t of 1.1 would not pass even a one-test bar of 1.96.
- Cost: 400 regressions of 41 observations, still seconds.

Screen 3 — OCMT: the same test, with a bar that rises with the crowd

The question it asks: *With 400 tests at once, which candidates are significant after allowing for the number of tests, and which were hidden behind the first picks?*

The rule (Chudik, Kapetanios and Pesaran, 2018: *one covariate at a time, multiple testing*).

1. **Stage 1.** Run the t-stat screen. Keep a candidate only if $|t|$ exceeds a bar that grows with the number of candidates p: with 400 tests at 5%, a bar of 1.96 would let about 20 useless series through by chance.
2. **Stage 2.** Put the stage-1 picks in as controls and test every remaining candidate again, with a higher bar. This recovers series masked by a stronger one; repeat until nothing new passes.

Critical value of $\lvert t \rvert$	one test	$p = 5$ (our example)	$p = 400$ (the panel)
Stage 1	1.96	2.58	**3.84**
Stage 2	–	2.84	4.52

$$c_p = \Phi^{-1}\left(1 - \frac{\alpha}{2\,p^{\delta}}\right), \qquad \alpha = 0.05,\ \delta = 1 \text{ (stage 1)},\ \delta^{*} = 1.5 \text{ (stage 2)}$$

Screen 3 by hand: two stages on five candidates

Stage 1, bar 2.58 for five candidates:

Candidate	$\lvert t \rvert$	verdict
Jobs searches	9.1	**in**
ATM withdrawals	5.3	**in**
Private credit	4.8	**in**
Real estate credit	3.6	**in**
Fuel imports	1.1	out

Stage 2, bar 2.84, fuel imports tested with the four picks as controls: $|t| = 0.17$. Out. The screen stops.

- OCMT returns a *set*, not just an order: four selected, one rejected. The tool turns it into a ranking (selected first, by their t at entry; the rest by their last t) so that it can propose groups of any size like the other screens.
- **The bar matters.** On the full panel the stage-1 bar is 3.84: real estate credit, at 3.6, would not make the first cut and would have to be rescued in stage 2.
- The payoff of stage 2 is the rescue of masked series: a candidate whose t is small on its own but large once a stronger series is controlled for. None here; on the panel it is why OCMT's list differs from the t-stat list after the first picks.

Screen 4 — LARS: whom would a regression hire first, if they compete?

The question it asks: *Letting all candidates compete for the same variation in GDP, in what order do they enter a regression?*

The rule (Efron, Hastie, Johnstone and Tibshirani, 2004). Start with every coefficient at zero and the residual equal to y minus its mean.

1. Find the candidate most correlated with the current residual. Move its coefficient slowly in that direction.
2. As the coefficient grows, the residual shrinks along that candidate until some other candidate's correlation catches up. The second one *enters*, and both coefficients then move along the direction that keeps them equally correlated with the residual (the "least angle").
3. Repeat until every candidate has entered. The **order of entry is the ranking**.

- A candidate that only repeats an earlier entrant never catches up: once jobs searches is in, the residual has little left for ATM withdrawals. That is the competition SIS lacks.
- The path is the LASSO path with a small change: LARS shows what a penalised regression would do at every strength of the penalty.
- The tool runs LARS on the panel, removes the entrants, runs it again on the rest, and so on (Chinn, Meunier and Stumpner, 2023), until the 400 are ordered.

Screen 4 by hand: the order of entry

Step 0: correlation of each (standardised) candidate with $y - \bar{y}$:

Candidate	corr	enters
Jobs searches	−0.70	**1st**
ATM withdrawals	−0.69	2nd
Private credit	0.51	3rd
Real estate credit	0.41	4th
Fuel imports	0.16	5th

On five candidates the order of entry coincides with the SIS order; the competition shows in the *coefficients* along the path, not in the order. On 400 candidates the two orders diverge quickly (next frame).

- Jobs searches enters first (largest correlation, 0.70). Its coefficient grows until ATM withdrawals, starting from 0.69, is equally correlated with what is left: a short first step, because the two started almost level.
- From then on the two move together. Because they are themselves correlated (0.66), each takes part of the other's role and both coefficients grow more slowly than they would alone.
- Private credit enters third, carrying information the pair does not have (its correlation with the pair is −0.02 and −0.48).
- **A note on scale.** LARS compares correlations only if the candidates are standardised. The textbook version standardises; the tool's call passes the panel as it is, so a series with very large swings can enter early for its scale rather than its signal. Flagged for review on 26 October 2026.

Screen 5 — SHAP: whom does a boosted-tree model actually use?

The question it asks: *Fit a flexible, non-linear model on all candidates; which ones does it rely on?*

The rule. Two steps.

1. Fit a **gradient boosted ensemble** of shallow trees (300 trees of depth 2, learning rate 0.05; see the trees deck) on GDP growth with all candidates as inputs. Shallow trees and a small step keep it from memorising 41 quarters.
2. For every training quarter and candidate, compute the **SHAP value**: the part of that quarter's prediction attributable to that candidate (Lundberg and Lee, 2017). Importance = **average absolute SHAP value** over the quarters. A candidate the trees never split on scores zero.

- **What a SHAP value is.** Each prediction is the sample mean plus one contribution per candidate, adding up exactly: the candidate's average marginal effect over every order in which the candidates could be added (the Shapley value), computed exactly for trees.
- **Why a non-linear screen.** The other five are linear at heart. SHAP can rank a series that matters only below a threshold, or only together with another.
- Competitive: two candidates with the same information share the splits, and both scores fall.

$$\hat{y}_t = \bar{y} + \sum_j \phi_{jt}, \qquad \mathrm{SHAP}(x_j) = \tfrac{1}{T}\sum_t |\phi_{jt}|$$

Screen 5 by hand: where each prediction comes from

The boosted ensemble fitted on the twelve quarters, and the contributions ϕ_{jt} for two of them (sample mean $\bar{y} = -1.17$):

Candidate	2016-Q4 ($y = 3.72$)	2018-Q3 ($y = -2.83$)	mean $\lvert\phi_j\rvert$ (all 12)
Jobs searches	3.88	0.23	**0.92**
Private credit	0.52	−1.09	0.69
Real estate credit	0.28	−0.42	0.49
ATM withdrawals	0.17	−0.33	0.13
Fuel imports	0.04	−0.05	0.02
Sum of contributions	4.89	−1.66	
Prediction $= \bar{y} +$ sum	3.72	−2.83	

With twelve points and 300 trees the ensemble reproduces every training quarter; the exercise shows the bookkeeping, not forecasting skill.

- 2016-Q4, the one strong quarter (+3.7): almost all of it is attributed to jobs searches, which fell by 6.6% that quarter, the largest drop in the window.
- 2018-Q3: the credit series carry the negative reading; jobs searches, flat at 0.05, contributes little.
- **The ranking.** Jobs searches first again, then the two credit series. ATM withdrawals, second for SIS and t-stat, drops to fourth: once the trees have split on jobs searches, the ATM series rarely improves a split. That is competition working on a correlated pair.
- Fuel imports is nearly never used.

Screen 6 — iBMA: in how many good models does the candidate appear?

The question it asks: *Across all plausible small regressions, how often is the candidate in the model, weighted by how well each model fits?*

The rule (Yeung, Bumgarner and Raftery, 2005: *iterated Bayesian model averaging*, with Raftery's BIC approximation).

1. **Pecking order.** Rank the candidates by their univariate R^2 with GDP growth.
2. **Window.** Take the first $w = 10$ and estimate *every* regression they can form: $2^{10} = 1{,}024$ models, each weighted by $\exp(-\mathrm{BIC}/2)$: a better fit raises the weight, every extra variable lowers it.
3. **Posterior inclusion probability (PIP).** The summed weight of the models that contain the candidate.
4. **Iterate.** Candidates with PIP below 0.05 leave the window, the next in the pecking order fill the seats; repeat until the list is exhausted. The ranking is by PIP.

- Competitive by construction: a duplicate sits in models that fit no better and pay the BIC penalty twice, so its PIP falls.
- On the panel: windows of ten, about 40 rounds of 1,024 regressions; seconds, not minutes.

$$\mathrm{BIC}_m = n \ln\!\left(\tfrac{\mathrm{RSS}_m}{n}\right) + k_m \ln n, \qquad w_m \propto e^{-\mathrm{BIC}_m/2}, \qquad \mathrm{PIP}(x_j) = \textstyle\sum_{m \ni j} w_m$$

Screen 6 by hand: a toy window of three candidates, eight models

Pecking order by univariate R^2: jobs searches 0.49, ATM withdrawals 0.47, private credit 0.26, real estate credit 0.17, fuel imports 0.03. A window of three (instead of ten) gives $2^3 = 8$ models; $n = 12$.

Model	variables	RSS	BIC	weight
1	(constant only)	62.9	22.4	0.003
2	jobs	32.2	16.8	0.047
3	ATM	33.3	17.2	0.039
4	jobs, ATM	26.6	17.0	0.043
5	private credit	46.8	21.3	0.005
6	jobs, private credit	16.9	**11.6**	**0.654**
7	ATM, private credit	30.7	18.7	0.018
8	jobs, ATM, private credit	16.9	14.0	0.190

PIP jobs $= 0.047 + 0.043 + 0.654 + 0.190$	**0.93**
PIP private credit $= 0.005 + 0.654 + 0.018 + 0.190$	0.87
PIP ATM $= 0.039 + 0.043 + 0.018 + 0.190$	0.29

- Model 6, jobs searches plus private credit, fits almost as well as the full model 8 (RSS 16.9 both) with one variable fewer, so the BIC penalty hands it two thirds of the weight.
- ATM withdrawals adds nothing once jobs searches is in (model 4 barely improves on model 2; model 8 does not improve on model 6). Its PIP collapses to 0.29, while SIS had it in a near tie for first.
- In the real screen, with PIP above 0.05 all three stay in the window and the next candidates join; a series that keeps losing to its neighbours is eventually swapped out.
- Weights: $w_m \propto \exp(-(\mathrm{BIC}_m - \mathrm{BIC}_{\min})/2)$, normalised to sum to one.

Part III — From six rankings to the pool

Six interviewers, six lists

The example: rank of each candidate under each screen (1 = best).

	SIS	t	OCMT	LARS	SHAP	iBMA
Jobs searches	1	1	1	1	1	1
ATM withdrawals	2	2	2	2	4	3
Private credit	3	3	3	3	2	2
RE credit	4	4	4	4	3	–
Fuel imports	5	5	5	5	5	–

iBMA: the three in the toy window. The one-at-a-time screens agree with each other; the competitive screens demote the duplicate (ATM) and promote the complement (credit).

The real run: rank of the same five among 400 candidates, X-13 identification run, rankings in force at 2020-Q2.

	SIS	t	OCMT	LARS	SHAP	iBMA
Jobs searches	4	–	–	73	2	**1**
Fuel imports	10	13	13	29	57	**2**
RE credit	7	4	4	–	24	**3**
Private credit	74	36	36	–	5	10
ATM withdrawals	5	9	9	–	–	**6**

"–": outside the top 99 (t = t-stat). All five are in the iBMA top 6 of that date, the group behind the watchlist's zero cluster; LARS has none in its first 28, SHAP two in its top 5.

The lesson of the deck: no screen is right on its own. The tool keeps all six and lets a tryout decide.

Before any ranking: three redundancy rules

The screens see cleaner applicants if the obvious duplicates are removed first. Three rules, applied to the quarterly panel before the six rankings (Jamaica method, 2026):

1. **Identities.** An aggregate and its components are one piece of information. From a typed list of identities (fiscal, external, credit aggregates) the components whose share of the aggregate is below 20% are dropped; the aggregate stays.
2. **The 0.8 prune.** Walking down each screen's ranking, a candidate is kept only if its correlation with every candidate already kept is below 0.8. The rankings stay nested and lose their near-copies. (Jobs searches and ATM withdrawals, at 0.66, both survive.)
3. **The 20-sigma guard.** A monthly series whose standardised growth exceeds 20 standard deviations of its own training distribution anywhere is treated as broken (a rate leaving zero, a reclassification): 53 series on the first Trinidad panel, 18 on the production panel of 26 October 2026. They stay as candidates for the penalised and tree estimators, which cope with outliers, but leave the within-quarter shock features and the breadth count.

After the rules: **400 candidates** ranked by each of the six screens, to a depth of 99.

The tryout: groups of the top K, scored on quarters the ranking never saw

1. **Groups.** For every screen and every size K from 5 to 99, the group is the top K of that screen's ranking: $6 \times 95 = 570$ groups.
2. **One cheap model per group.** Standardise the group, take its first principal component, regress GDP growth on it. The same model for every group, so only the group is being judged.
3. **Expanding-window cross-validation.** Train on the first 16 quarters, forecast the next 4; add 2 quarters, repeat: 17 folds over the training window. Score = average MSE of the forecasts.
4. **The benchmark.** The same model on *all* 400 candidates. A group **beats the benchmark** if its ratio of CV MSE to the benchmark's is below 1.

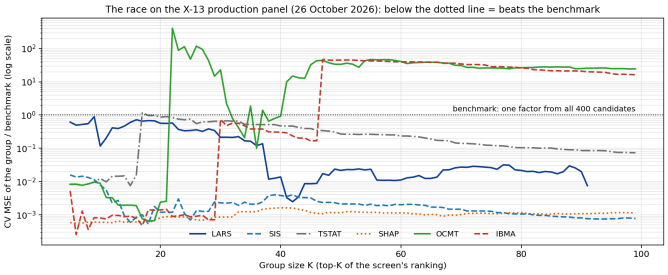
Only the training window is used. The test quarters of the pipeline (2023-Q2 onward) never enter the race.

Why this scoring and not the full estimator set? Because the race must be cheap: 570 groups $\times$ 17 folds is 9,690 small regressions, seconds. The expensive estimators (ridge, elastic net, LASSO, random forests, boosting, DT4, LSTM) are fitted later, only on the groups that survive.

Why one factor? On 41 quarters the Bai-Ng criterion selects one factor for every group; more factors would reward large groups for fitting noise.

The race is rolling. In the test window the rankings are refitted every quarter with the data then available, so a group named "SIS top 17" is a different list in 2024-Q1 and 2024-Q2. Two consecutive SIS top-17 lists shared only 4 of 17 variables in the identification run.

The race on the Trinidad and Tobago panel (production run, 26 October 2026)

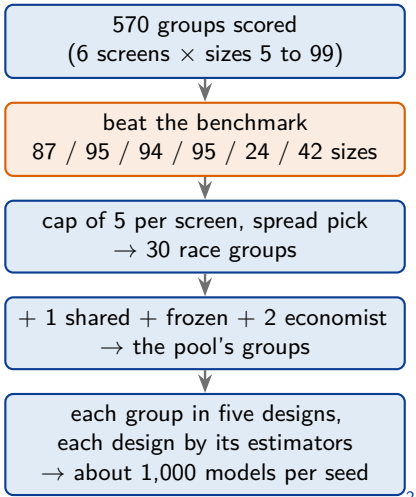


- The all-candidates benchmark is very poor on this panel (one factor of 400 noisy growth rates), so the bar is low: SIS, SHAP and t-stat beat it at 94 or 95 of the 95 sizes, LARS at 87.
- OCMT and iBMA are different: their orderings are only meaningful up to the set they select (24 and 42 sizes beat the benchmark); past that point the "ranking" is the leftovers, and the groups blow up.
- Best sizes: iBMA 6, t-stat 15, SIS, SHAP and OCMT 18, LARS 42. Small groups win on 41 quarters.

Log scale; below the dotted line = beats the benchmark.

From 570 groups to the pool: the cap, the spread pick and the extra groups

- **Cap of five per screen.** Each screen with at least one winning size keeps five groups. Without the cap the clean panel returned 60 groups and 300 specifications.
- **Spread pick, not best five.** The winning sizes are cut into five equal bins and the best group of each bin is kept, so the pool holds small and large groups of every screen, not five neighbours of the same size.
- **Consensus groups.** Variables in the largest winning group of every winning screen ("common") or of all but one ("shared") form two extra groups. On 26 October 2026 the common group was empty and the shared group had six variables, among them fuel imports.
- **Frozen groups.** The lists in force at COVID (2020-Q2) that read the crash, carried as static groups so the watchlist does not move with the rolling race (pipeline deck, Part IV).
- **Economist groups.** Two groups typed in from judgement (energy, activity), estimated beside the race.



How much do the six lists overlap? Not much

Share of common variables between the top-60 lists of two screens, production run of 26 October 2026:

	LARS	SIS	t	SHAP	OCMT	iBMA
LARS	1	0.15	0.30	0.23	0.33	0.13
SIS	0.15	1	**0.42**	0.28	0.17	**0.42**
t-stat	0.30	0.42	1	0.32	0.30	0.38
SHAP	0.23	0.28	0.32	1	0.25	0.27
OCMT	0.33	0.17	0.30	0.25	1	0.27
iBMA	0.13	0.42	0.38	0.27	0.27	1

- Two screens share at most 42 of their top 60, usually 15 to 30. Six interviewers, six genuinely different short lists.
- The closest pairs are the ones that start from the same statistic: SIS and iBMA (iBMA's pecking order is the univariate R^2, the square of the SIS correlation), SIS and t-stat.
- LARS is the loner (0.13 to 0.33): the only screen whose order is decided by competition at every step, and the one most exposed to scale.
- This is why the pool is built from all six rather than from a vote: a vote would keep the 15% they agree on and lose the rest, and the race shows that each screen's groups carry forecasting value of their own.

What to remember

- **Pre-selection is a ranking problem, not an estimation problem.** With 400 candidates and 41 quarters nothing can be estimated on everything; six cheap screens each propose an order, and a cheap tryout decides which short lists go on to the expensive estimators.
- **One-at-a-time screens count duplicates twice.** SIS, t-stat and OCMT interview alone and ranked jobs searches and ATM withdrawals, correlated at 0.66, first and second. The competitive screens (LARS, SHAP, iBMA) demoted the duplicate and promoted private credit, which carried new information.
- **Each screen asks a different question**, and the answers overlap by 15 to 40%. Keeping all six, with a cap and a spread of sizes, is a hedge against choosing the wrong question.
- **The race only sees the training window.** Groups are scored on quarters their ranking never used, and the pipeline's test quarters never enter the race; in the test window the rankings roll forward with the data.
- **The screens hand over groups, not forecasts.** Every surviving group is then estimated in five designs by up to seven estimators, three seeds each, and the pool, not any single group, produces the regime-aware nowcast.

References

- Bai, J. and S. Ng (2008). Forecasting economic time series using targeted predictors. *Journal of Econometrics* 146(2), 304–317.
- Chinn, M. D., B. Meunier and S. Stumpner (2023). Nowcasting world trade with machine learning: a three-step approach. NBER Working Paper 31419. (The four pre-selection methods compared: LARS, SIS, t-stat, iBMA.)
- Chudik, A., G. Kapetanios and M. H. Pesaran (2018). A one covariate at a time, multiple testing approach to variable selection in high-dimensional linear regression models. *Econometrica* 86(4), 1479–1512.
- Efron, B., T. Hastie, I. Johnstone and R. Tibshirani (2004). Least angle regression. *Annals of Statistics* 32(2), 407–499.
- Fan, J. and J. Lv (2008). Sure independence screening for ultrahigh dimensional feature space. *Journal of the Royal Statistical Society B* 70(5), 849–911.
- Jurado, K., S. C. Ludvigson and S. Ng (2015). Measuring uncertainty. *American Economic Review* 105(3), 1177–1216.
- Lundberg, S. M. and S.-I. Lee (2017). A unified approach to interpreting model predictions. *NeurIPS* 30.
- Raftery, A. E. (1995). Bayesian model selection in social research. *Sociological Methodology* 25, 111–163.
- Yeung, K. Y., R. E. Bumgarner and A. E. Raftery (2005). Bayesian model averaging: development of an improved multi-class, gene selection and classification tool for microarray data. *Bioinformatics* 21(10), 2394–2402.